Solve with us 4 - Not Graded

This assignment will not be graded and is only for practice.
Key points: 1Key points: 1

- The span of a set SS (of vectors) is defined as the set of all finite linear combinations of elements (vectors) of SS and denoted by $Span(S)$Span(S) i.e.,
  $Span(S) = \{\sum_{i=1}^{n} a_i v_i \in V \mid a_1, a_2, ..., a_n \in R$ and $v_1, v_2, ..., v_n \in S\}$Span(S) = {∑i=1n aivi ∈ V ∣ a1, a2, ..., an ∈ R and v1, v2, ..., vn ∈ S}.
- Let VV be a vector space. A set $S \subseteq V$S ⊆ V is spanning set for VV if $Span(S) = V$Span(S) = V .
- Algorithm to obtain a spanning set for a non-zero vector space V:
  Algorithm to obtain a spanning set for a non-zero vector space V:
  i) Start with the empty set $S_0 = \phi$S0 = ϕ.
  ii) Add any non-zero vector $v_1 \in V$v1 ∈ V to S_0S0 and define it as S_1S1. So $S_1 = S_0 \cup \{v_1\}$S1 = S0 ∪ {v1}.
  iii) If $Span(S_1) = V$Span(S1) = V , then we are done. Otherwise choose $v_2 \in V \setminus span(S_1)$v2 ∈ V ∖ span(S1) and add it to the set and call it S_2S2. So $S_2 = S_1 \cup \{v_2\}$S2 = S1 ∪ {v2}.
  iv) If $Span(S_2) = V$Span(S2) = V , then we are done. Otherwise, choose $v_3 \in V \setminus Span(S_2)$v3 ∈ V ∖ Span(S2), and construct $S_3 = S_2 \cup \{v_3\}$S3 = S2 ∪ {v3}.
  v) Repeat the process until $Span(S_n) = V$Span(Sn) = V for some nn.

1 point

Let $S = \{(1, 2), (2, 3), (1, -1)\} \subseteq$ S = {(1, 2), (2, 3), (1, −1)} ⊆ R^2R2. Which of the following is (are) elements of $Span(S)$Span(S)?

- ☐ 5(1,2) + 8(1,-1)
- ☐ (0,0)
- ☐ $2(1, 2) - 3(2, 3) + 5(1, -1)$2(1, 2) − 3(2, 3) + 5(1, −1)
- ☐ $(2, 1) - 2(3, 2) - 4(-1, 1)$(2, 1) − 2(3, 2) − 4(−1, 1)

No, the answer is incorrect.
Score: 0
Accepted Answers:
5(1,2) + 8(1,-1)
(0,0)
2(1, 2) − 3(2, 3) + 5(1, −1)2(1, 2) − 3(2, 3) + 5(1, −1)
(2, 1) − 2(3, 2) − 4(−1, 1)(2, 1) − 2(3, 2) − 4(−1, 1)

1 point

Let $S = \{(1, 1, -1), (-1, 1, 1)\} \subset R^3$ S = {(1, 1, −1), (−1, 1, 1)} ⊂ R3. Which of the following is (are) elements of $Span(S)$Span(S)?

- ☐ 2(1, 1, -1)+ (-1, 1, 1)
- ☐ (8, 8, -8)
- ☐ (0, 2,0)
- ☐ (1,1,-2)

No, the answer is incorrect.
Score: 0
Accepted Answers:
2(1, 1, -1)+ (-1, 1, 1)

(8, 8, -8)
(0, 2,0)
1 point
Let S = S = {(-1, 1, 1), (0, $\frac{1}{2}$21 ,0), (0, 1, -2)} $\subset R^3 \subset$ R3. Which of the following option(s) is (are) true?

[Hint:Hint: Form the matrix using the vectors as the columns and calculate the determinant of the matrix.]

- [ ] SS is a linearly dependent set.
- [ ] SS is a linearly independent set.
- [ ] SS is a basis of the vector space R^3 R3.
- [ ] None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
SS is a linearly independent set.
SS is a basis of the vector space R^3 R3.

1 point
Consider a set of vectors S = {(1, 2), (− 1, 2), (3, 1), (− 3, 1)}S = {(1, 2), (−1, 2), (3, 1), (−3, 1)} in R^2 R2. Choose the set of correct options.

- [ ] The set {(1, 2), (− 1, 2), (3, 1)}{(1, 2), (−1, 2), (3, 1)} is a basis of R^2 R2.
- [ ] The set {(− 1, 2), (3, 1)}{(−1, 2), (3, 1)} is a basis of R^2 R2.
- [ ] The set {(3, 1), (− 3, 1)}{(3, 1), (−3, 1)} is a basis of R^2 R2.
- [ ] None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The set {(− 1, 2), (3, 1)}{(−1, 2), (3, 1)} is a basis of R^2 R2.
The set {(3, 1), (− 3, 1)}{(3, 1), (−3, 1)} is a basis of R^2 R2.

Key points: 2Key points: 2

- The following conditions are equivalent for a set to be a basis of a vector space:
  - Maximal linearly independent set : A linearly independent set such that if any vector is appended to it, the new set is no longer linearly independent.
  - Minimal spanning set : A spanning set such that if any vector is removed from the set, then the set will not be a spanning set.

1 point
Which of the following sets are maximal linearly independent sets in R^2 R2?

- [ ] {(1, 0)}{(1, 0)}
- [ ] {(1, 2), (2, 4)}{(1, 2), (2, 4)}

☐ $\{(1,2),(1,1)\}$ {(1, 2), (1, 1)}

☐ $\{(1,2),(1,1),(2,1)\}$ {(1, 2), (1, 1), (2, 1)}

☐ $\{(1,0),(0,1)\}$ {(1, 0), (0, 1)}

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1,2),(1,1)\}$ {(1, 2), (1, 1)}

$\{(1,0),(0,1)\}$ {(1, 0), (0, 1)}

1 point

Which of the following sets are minimal spanning sets of R^3 R3?

☐ $\{(1,0,0),(0,1,0)\}$ {(1, 0, 0), (0, 1, 0)}

☐ $\{(1,0,0),(1,1,0),(1,1,1)\}$ {(1, 0, 0), (1, 1, 0), (1, 1, 1)}

☐ $\{(1,0,1),(0,1,1),(1,1,0)\}$ {(1, 0, 1), (0, 1, 1), (1, 1, 0)}

☐ $\{(1,0,1),(0,1,1),(1,1,2)\}$ {(1, 0, 1), (0, 1, 1), (1, 1, 2)}

☐ $\{(1,0,0),(0,1,0),(0,0,1),(1,1,1)\}$ {(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)}

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1,0,0),(1,1,0),(1,1,1)\}$ {(1, 0, 0), (1, 1, 0), (1, 1, 1)}

$\{(1,0,1),(0,1,1),(1,1,0)\}$ {(1, 0, 1), (0, 1, 1), (1, 1, 0)}

Key Points: 3**Key Points: 3**

- **Basis:** A basis of a vector space VV is a linearly independent subset of VV that spans VV .
- Basis of a vector space:
  - Maximal linearly independent set.
  - Minimal spanning set.
- How to find a basis of a subspace of R^n Rn?
  - Suppose a subspace VV of R^n Rn is defined by some constraints, as follows:

$V = \{(x_1, x_2, \ldots, x_n) \mid f_1(x_1, x_2, \ldots, x_n) = 0, f_2(x_1, x_2, \ldots, x_n) = 0, \ldots, f_m(x_1, x_2, \ldots, x_n)$
V = {(x1, x2, … , xn) ∣ f1(x1, x2, … , xn) = 0, f2(x1, x2, … , xn) = 0, … ,fm(x1, x2, … , xn) = 0} ⊆ Rn

Where f_i fi's are linear equations in general.

- **Step 1**: By solving f_i fi's we have to try to find out the dependent and independent variables. Suppose there are kk independent variables and there are $k - n$k − n dependent variables.

- **Step 2**: Choose the first independent variable, let it be x_1 x1 and assign the value 1

to that, and assign 0 to all the other independent variables.

- **Step 3**: Find out the values of all the dependent variables with respect to the assignment given in Step 2 and call that vector to be v_1 v1.
- **Step 4**: Carry out this process for each independent variable, to get the vectors $v_2, \ldots, v_k$ v2, … , vk.

The set of vectors $\{v_1, v_2, \ldots, v_k\}${v1, v2, … , vk} is a linearly independent set and this is a maximal linearly independent set in VV , hence forms a basis of VV . Hence, dimension of VV is kk in this case.

1 point

Let S = S ={(-1, 1, 1), (0, $\frac{1}{2}$21 ,0), (0, 1, -2)} $\subset R^3 \subset$ R3. Which of the following option(s) is (are) true?

Hint:Hint: Form the matrix using the vectors as the columns and calculate the determinant of the matrix.

- ☐ SS is a linearly dependent set.
- ☐ SS is a linearly independent set.
- ☐ SS is a basis of the vector space R^3 R3.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

SS is a linearly independent set.

SS is a basis of the vector space R^3 R3.

1 point

Consider a set of vectors S = {(1, 2), (− 1, 2), (3, 1), (− 3, 1)}S = {(1, 2), (−1, 2), (3, 1), (−3, 1)} in R^2 R2. Choose the set of correct options.

- ☐ The set { (1, 2) ,(-1, 2), (3, 1)} is a basis of R^2 R2.
- ☐ The set { (-1, 2), (3, 1) } is a basis of R^2 R2.
- ☐ The set { (3, 1), (-3, 1) } is a basis of R^2 R2.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set { (-1, 2), (3, 1) } is a basis of R^2 R2.

The set { (3, 1), (-3, 1) } is a basis of R^2 R2.

1 point

Let VV be a vector space which is defined as follows:

$V = \{(x, y, z) \mid 2x + 3y = 0$ and $y + 5z = 0\} \subseteq R^3$V = {(x, y, z) ∣ 2x + 3y = 0 and y + 5z = 0} ⊆ R3 with usual addition and scalar multiplication. Which of the following sets form bases of VV ? (More than one option may be correct)

[Hint: Two constraints are given here, $f_1(x, y, z) = 2x + 3y = 0$f1(x, y, z) = 2x + 3y = 0 and $f_2(x, y, z) = y + 5z = 0$f2(x, y, z) = y + 5z = 0. Observe that zz is the independent variable and xx and yy are the dependent variables.]

☐ $\{(2,3,0),(0,1,5)\}$ {(2, 3, 0), (0, 1, 5)}

☐ $\{(3,-2,0),(0,5,-1)\}$ {(3, −2, 0), (0, 5, −1)}

☐ $\{(\frac{15}{2},-5,1)\}$ {(215, −5, 1)}

☐ $\{(15,-10,2)\}$ {(15, −10, 2)}

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(\frac{15}{2},-5,1)\}$ {(215, −5, 1)}
$\{(15,-10,2)\}$ {(15, −10, 2)}

1 point
Let V V be a vector space which is defined as follows: $V = \{(x, y, z, w) \mid x + z = y + w\} \subseteq R^4$ V = {(x, y, z, w) | x + z = y + w} ⊆ R4 with usual addition and scalar multiplication. Which of the following set forms a basis of V V ?

[Hint: The constraint given here is $x + z = y + w$x + z = y + w, which can be written as $x - y + z - w = 0$x − y + z − w = 0. As there are 4 variables and 1 equation, the number of independent variables is 3, where as the number of dependent variable is 1. If you check the first vector of each option, only Option 2 and Option 4 satisfying the constraint, and in that vector $x = y = 1$x = y = 1. Start with taking yy as the dependent variable, and apply the method mentioned above.]

○ $\{(1,0,0,0),(0,1,0,0),(0,0,0,1)\}$ {(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 0, 1)}.

○ $\{(1,1,0,0),(0,1,-1,0),(0,-1,0,1)\}$ {(1, 1, 0, 0), (0, 1, −1, 0), (0, −1, 0, 1)}.

○ $\{(1,-1,0,0),(0,1,1,0),(0,-1,0,1)\}$ {(1, −1, 0, 0), (0, 1, 1, 0), (0, −1, 0, 1)}.

○ $\{(1,1,0,0),(0,1,1,0),(0,-1,0,1)\}$ {(1, 1, 0, 0), (0, 1, 1, 0), (0, −1, 0, 1)}.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(1,1,0,0),(0,1,1,0),(0,-1,0,1)\}$ {(1, 1, 0, 0), (0, 1, 1, 0), (0, −1, 0, 1)}.

Find the dimension of the vector space $V = \{(x, y, z, w) \mid x + y = z + w, x + w = y + z, \text{ and } x, y, z, w \in R\}$V = {(x, y, z, w) | x + y = z + w, x + w = y + z, and x, y, z, w ∈ R}. [Hint: Two constraints are given here $f_1(x, y, z, w) = x + y - z - w = 0$f1(x, y, z, w) = x + y − z − w = 0 and $f_2(x, y, z, w) = x - y - z + w = 0$f2(x, y, z, w) = x − y − z + w = 0. Try to find out the number of independent variables.]

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

Key Points: 4**Key Points: 4**

- How to find a basis of a subspace VV of $M_{m \times n}(R)$Mm×n(R):

- **Step 1**: Consider each element of a $m \times n$m × n matrix as a variable. So there are mnmn number of variables.

- **Step 2**: Using the constraints given as the definition of the subspace separate out the independent and the dependent variables. Suppose there are kk independent variables and $mn - k$mn − k dependent variables.

- **Step 3**: Define a matrix M_1 M1 by assigning 1 to one independent variable and 0 to all the other independent variables, together with finding out the values of all the dependent variables.

- **Step 4**: Repeat this process for all the independent variables to get the vectors $M_2, M_3, \ldots, M_k$ M2, M3, … , Mk.

The set of matrices $\{M_1, M_2, \ldots, M_k\}${M1, M2, … , Mk} forms a basis of VV . Hence the dimension of VV is kk in this case.

1 point

Let $V = \{A \mid A \in M_{3 \times 3}(R), A$ is a diagonal matrix $\}$V = {A ∣ A ∈ M3×3 (R), A is a diagonal matrix }. Which of the following sets form a basis of VV ?

[Hint: $a_{ij} = 0$aij = 0 if $i \neq j$i = j. Thus there are 6 constraints and these are the dependent variables while the diagonal entries are the 3 independent elements.]

○ $\{\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}\}$

○ $\{\begin{bmatrix}1&0&0\\0&1&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&0\\0&0&0\\0&0&1\end{bmatrix}\}$

○ $\{\begin{bmatrix}1&0&0\\0&0&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&0\\0&1&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&0\\0&0&0\\0&0&1\end{bmatrix}\}$

○ $\{\begin{bmatrix}1&0&0\\0&0&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&0\\0&1&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&0\\0&0&0\\0&0&1\end{bmatrix}\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{\begin{bmatrix}1&0&0\\0&0&0\\0&0&0\end{bmatrix},\begin{bmatrix}0&0&0\\0&1&0\\0&0&0\end{bmatrix},\begin{bmatrix}0&0&0\\0&0&0\\0&0&1\end{bmatrix}\}$ $\left\{\begin{bmatrix}1&0&0&0&0&0&0&0&0\end{bmatrix},\begin{bmatrix}0&0&0&0&1&0&0&0&0\end{bmatrix},\begin{bmatrix}0&0&0&0&0&0&0&0&1\end{bmatrix}\right\}$

Let $V = \{A \mid A \in M_{3\times 3}(R)$, AV = {A | A ∈ M3×3(R), A is a diagonal matrix and sum of the diagonal elements are 0 }}. What will be the dimension of VV ?

[Hint: The general form a diagonal matrix of order 3 is $\begin{bmatrix}a_1&0&0\\0&a_2&0\\0&0&a_3\end{bmatrix}$ $\begin{bmatrix}a1&0&0&0&a2&0&0&0&a3\end{bmatrix}$. Hence there are three variables, a_1 a1, a_2 a2, a_3 a3. Now there is one more constraint, which says that the sum of the diagonal elements is 00. Hence we have $a_1 + a_2 + a_3$ = 0a1 + a2 + a3 = 0. So we can choose a_3 a3 to be the dependent variable and a_1 a1 and a_2 a2 to be the independent variables, as $a_3 = -a_1 - a_2$ a3 = −a1 − a2.]

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

Key points: 5 Key points: 5

- Rank of a matrix:
  - Maximum number of linearly independent column vectors (or row vectors) in a matrix.

  NOTE: Rank of a matrix is always less than or equal to the number of columns (or rows).

- How to find rank of a matrix A:
  - Reduce the matrix to row echelon form.
  - Find the number of non zero rows in the reduced matrix which will be the rank of the matrix A.

1 point
Consider the following two matrices:

P = $\begin{bmatrix}1&0&3\\2&-2&0\\0&1&-1\\0&2&1\end{bmatrix}$ $\begin{bmatrix}1&2&0&0\\0&-2&1&2\\3&0&-1&1\end{bmatrix}$; Q = $\begin{bmatrix}1&1&-2&0\\1&0&2&0\\0&-1&0&4\end{bmatrix}$ $\begin{bmatrix}1&1&0&1&0&-1&-2&2&0&0&0&4\end{bmatrix}$

Which of the following statements is (are) TRUE? [Hint: Try to reduce the matrices to row echelon form and find the number of non zero rows]

- ☐ Rank(P) = 4
- ☐ Rank(P) = Rank(Q)
- ☐ Rank(QP) = 4
- ☐ Rank(PQ) = Rank(QP)

No, the answer is incorrect.
Score: 0
Accepted Answers:
Rank(P) = Rank(Q)
Rank(PQ) = Rank(QP)

Check Answers